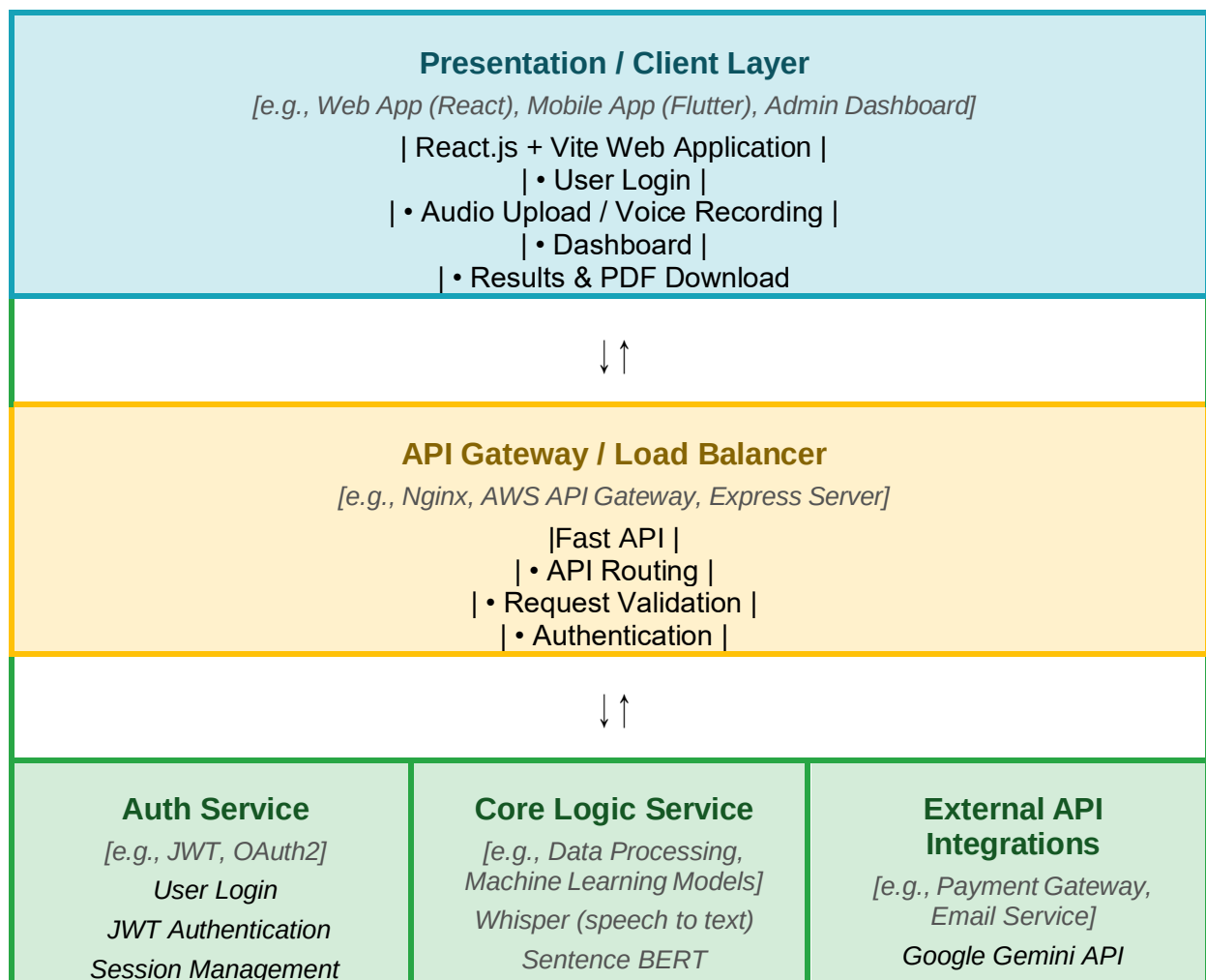


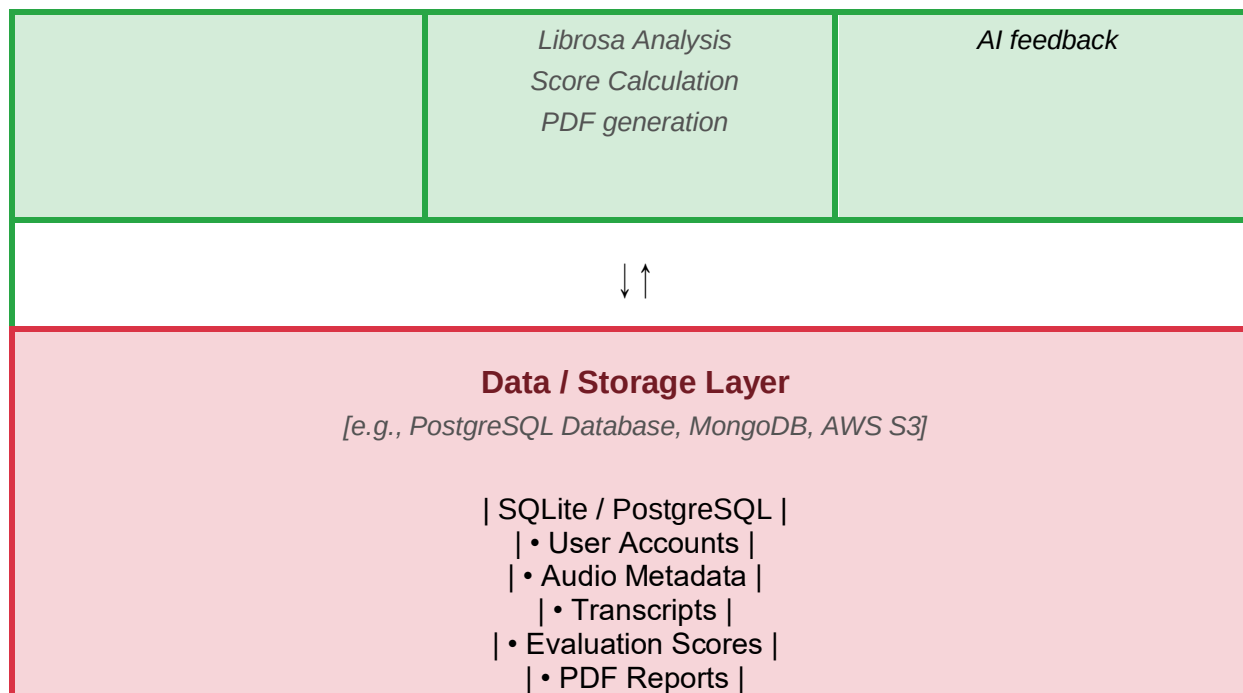
Solution Architecture

Date	26 June 2026
Team ID	SWTID-2026-9097
Project Name	Voice--Based--Concept--Understanding--Analyzer
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.





Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the user interface for authentication, audio recording/upload, viewing transcripts, evaluation scores, AI feedback, and downloading PDF reports.	React.js, Vite, Tailwind CSS, HTML5, CSS3
API Gateway	Receives requests from the frontend, validates input, manages authentication, and routes requests to AI processing services.	FastAPI, REST API
Authentication Service	Authenticates users, manages sessions, and restricts access to authorized users only.	JWT, FastAPI Security
Core Logic Service	Performs speech-to-text conversion, semantic similarity analysis, speech feature extraction, score calculation, report generation, and workflow orchestration.	Python, Whisper, Sentence-BERT, Librosa, NumPy, ReportLab
External API Integrations	Generates personalized AI feedback and improvement suggestions based on evaluation results.	Google Gemini API
Database / Storage Layer	Stores user accounts, transcripts, evaluation history, scores, and generated PDF reports securely.	SQLite (Development), PostgreSQL (Production)